

Answer **all** questions in the spaces provided.

**0 1 . 1**

**Table 1** shows features of a bacterium and the human immunodeficiency virus (HIV) particle.

Complete **Table 1** by putting a tick (✓) where a feature is present.

**[2 marks]**

**Table 1**

| Feature          | Bacterium | Human immunodeficiency virus (HIV) particle |
|------------------|-----------|---------------------------------------------|
| RNA              |           |                                             |
| Cell wall        |           |                                             |
| Enzyme molecules |           |                                             |
| Capsid           |           |                                             |

**0 1 . 2**

When HIV infects a human cell, the following events occur.

- A single-stranded length of HIV DNA is made.
- The human cell then makes a complementary strand to the HIV DNA.

The complementary strand is made in the same way as a new complementary strand is made during semi-conservative replication of human DNA.

Describe how the complementary strand of HIV DNA is made.

**[3 marks]**

---

---

---

---

---

---

---

---

---

---



|   |   |
|---|---|
| 0 | 1 |
|---|---|

 . 

|   |
|---|
| 3 |
|---|

Contrast the structures of DNA and mRNA molecules to give **three** differences.**[3 marks]**

1 \_\_\_\_\_

\_\_\_\_\_

2 \_\_\_\_\_

\_\_\_\_\_

3 \_\_\_\_\_

\_\_\_\_\_

**Turn over for the next question**